

RISP International Workshop on Nonlinear Circuits, Communications and Signal Processing 2018

**Hilton Waikiki Beach Hotel, Honolulu, Hawaii
March 4 - 7, 2018**

Effective Feature Selection for Damaged Buildings Using Post-Earthquake Satellite Image with Machine Learning

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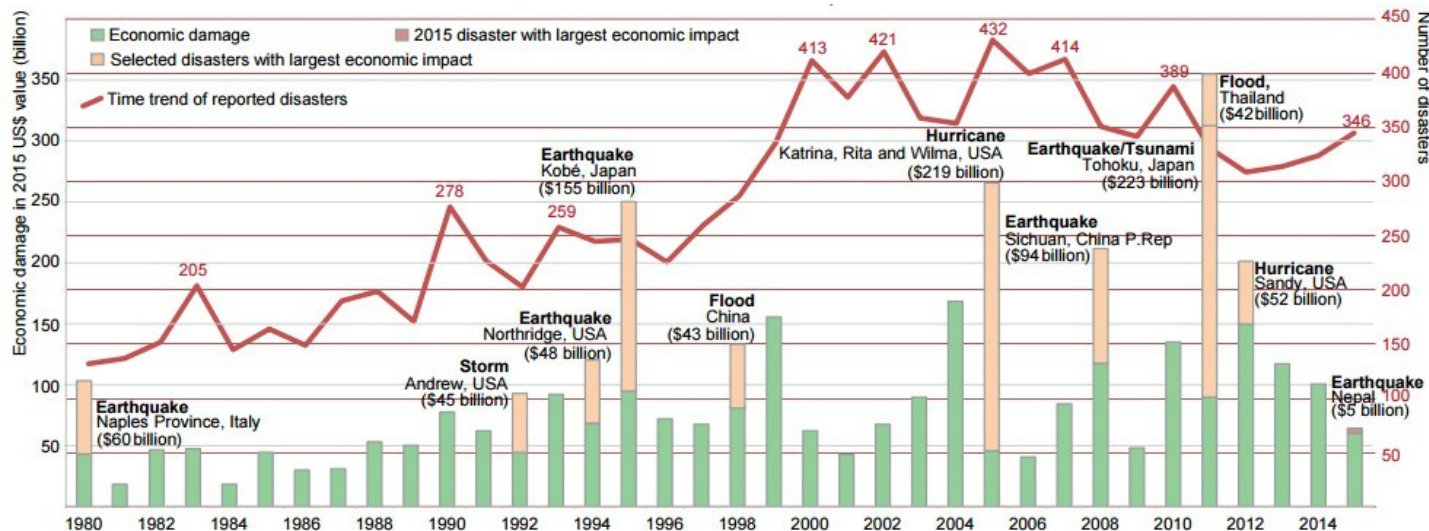
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Hidehiro Kanemitsu

United Nations Office for Disaster Risk Reduction (UN/ISDR)

- Its mandate was expanded to serve as the focal point in the United Nations system to ensure coordination and synergies among disaster risk reduction activities of the United Nations system and regional organizations and activities in socio-economic and humanitarian fields

Annual reported economic damages and time trend from disasters: 1980-2015



Major earthquake occurrence in recent years

- 2016/10 — Central Italy earthquakes(M6.5)
- 2016/04 — Ecuador coast (M7.8)
- 2016/04 — Kumamoto, Japan (M7.1)
- 2016/03 — Near Indonesia (M7.9)
- 2015/09 — Earphel, Chile (M8.3)
- 2015/04 — Nepal (M7.8)
- 2014/06 — Aleutian Archipelago (M8.0)
- 2014/04 — Iquique, Chile (M8.2)
- 2013/04 — Sichuan Province (M7.0)
- 2012/04 — Sumatra, Indonesia (M8.7)
- 2011/03 — East Japan great earthquake (M9.0)
- 2010/02 — Maure, Chile (M8.8)
- 2010/01 — Haiti (M7.0)**



Central Italy earthquakes(M6.5)



Sichuan Province (M7.0), China

Background

- **If disasters occur**, the earlier the rescue and relief activities are carried out, the more damage can be suppressed.
- To response immediately, voluntary disaster prevention and volunteer activities by organization that know disaster area well are necessary.
- However, in many cases firefighting places and police stations in the area are also damaged, so activities such as relief can not be smoothly carried out in the initial stage.
- Therefore, relief activities from other areas are also important but they are unfamiliar with that area.
- **A map** showing the disaster situation is needed so that volunteer organizations from other areas can work smoothly.(Not only populated areas, but small villages in mountainous areas)

Object

- The purpose of this study
 - to create data to support near real-time relief and rescue efforts in a disaster affected area by locating damage in a short period of time using only post-disaster satellite images of the area.
1. only high-resolution satellite images after disaster
 2. eliminating the use of complex algorithms and auxiliary data

Study area

- On January 12th 2010, Haiti was hit by a magnitude 7.0 M_w earthquake with an epicenter about 13 km deep and 25 km west of Port-au-Prince, Haiti's capital.
- Most buildings in Haiti are built of reinforced concrete frames with walls of piled-up unglazed bricks. Due to the strong quake, many of the buildings including the parliament house experienced so-called “pancake crash,” in which buildings were flattened with ceilings and floors on top of one another in layers.



Data (GeoEye-1)

Before Earthquake
Jul 27, 2009



After Earthquake
Jan 13, 2010



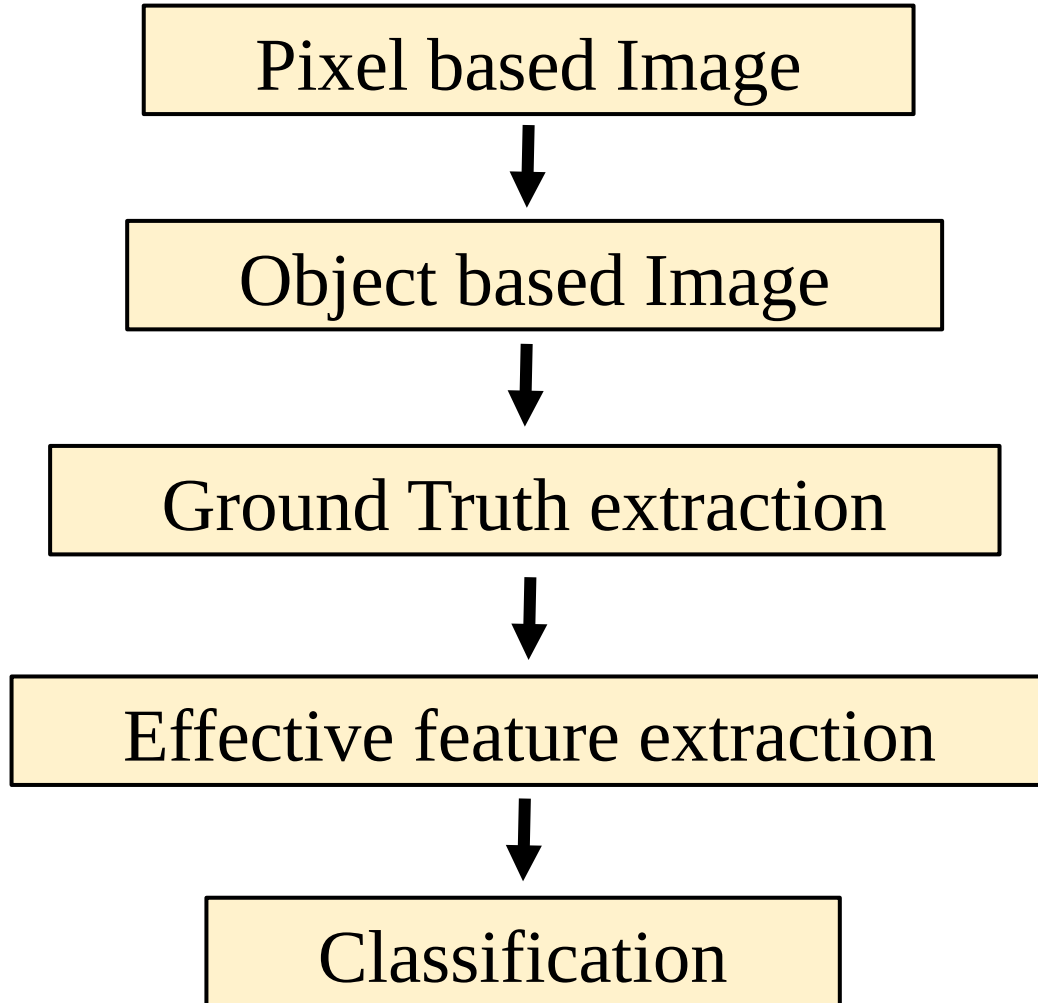
Launch Date September 6, 2008

Resolution 0.46 m / panchromatic
1.84 m / multispectral

Spectral Range Panchromatic: 450 - 800 nm
Blue: 450 - 510 nm
Green: 510 - 580 nm
Red: 655 - 690 nm
Near Infra Red: 780 - 920 nm

Equator Crossing Time 10:30am

Method



Segmentation

Gray Level Co-occurrence Matrix (GLCM)
Haralick feature
Hierarchical region segmentation
Object figure anlaysis

Tree
Road : Affective Building
Building

Stepwise Method

<< Haralick Texture Feature >>

$$\sum_{i,j=0}^{N-1} \frac{P_{i,j}}{1 + (i - j)^2}$$

Homogeneity

$$\sum_{i,j=0}^{N-1} P_{i,j}^2$$

Ang.2nd moment

$$P_{i,j} = \frac{V_{i,j}}{\sum_{i,j=0}^{N-1} V_{i,j}}$$

$$\sum_{i,j=0}^{N-1} P_{i,j} (i - j)^2$$

Contrast

$$u_{i,j} = \frac{\sum_{i,j=0}^{N-1} P_{i,j}}{N^2}$$

Mean

$$\sum_{i,j=0}^{N-1} P_{i,j} |i - j|$$

Dissimilarity

$$\sigma_{i,j}^2 = \sum_{i,j=0}^{N-1} P_{i,j} (i, j - u_{i,j})^2$$

Std dev

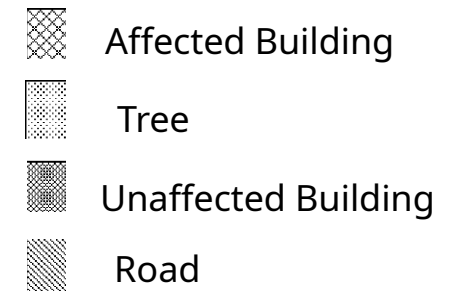
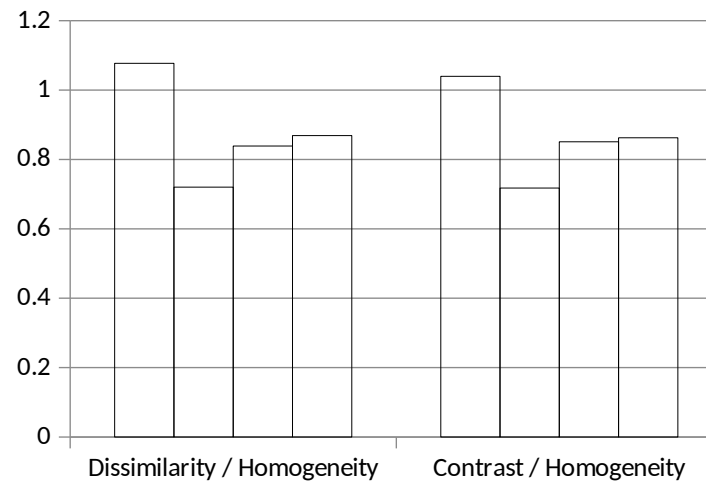
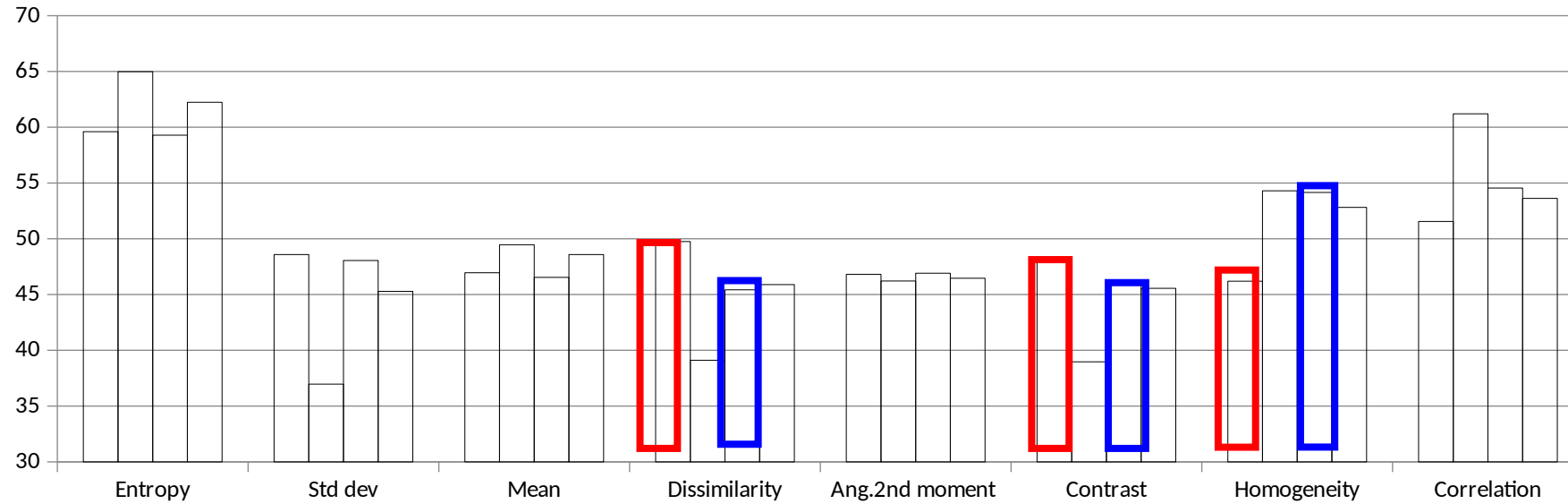
$$\sum_{i,j=0}^{N-1} P_{i,j} (-\ln P_{i,j})$$

Entropy

$$\sum_{i,j=0}^{N-1} P_{i,j} \left[\frac{(i - u_i)(j - u_j)}{\sqrt{(\sigma_i^2)(\sigma_j^2)}} \right]$$

Correlation

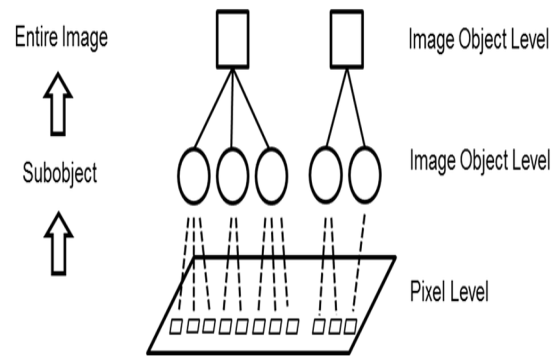
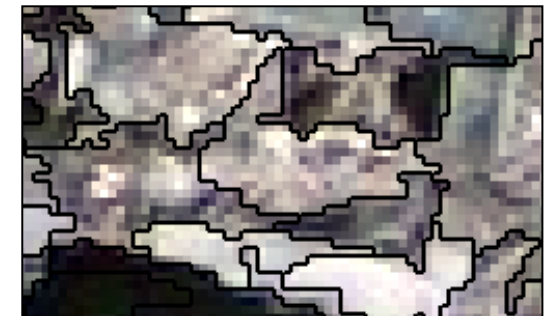
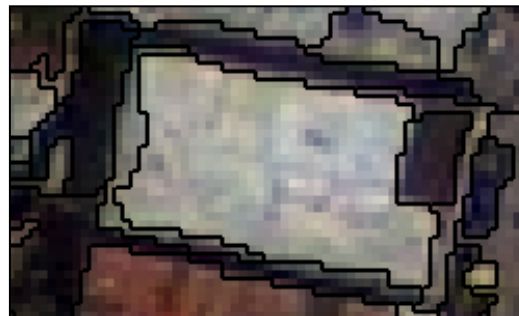
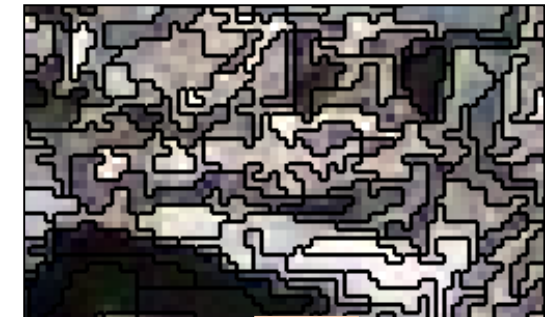
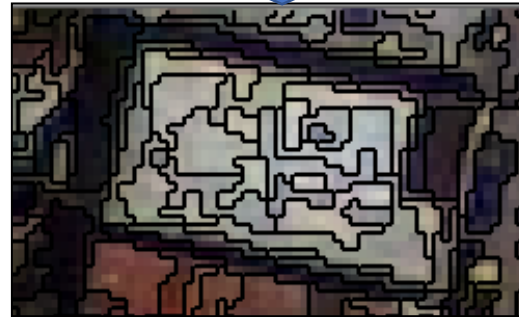
Haralick Texture Feature



<< Hierarchical region segmentation >

Before Earthquake

After Earthquake

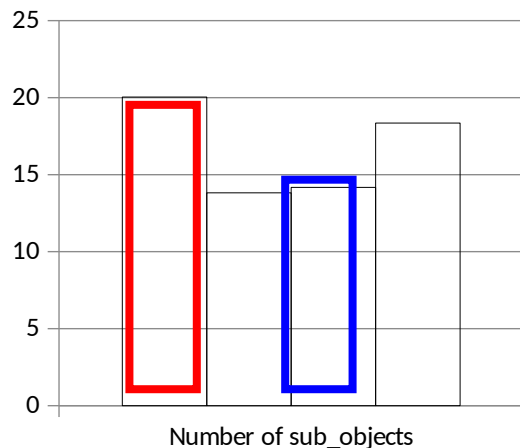


1st Segmentation
Scale Parameter : 15
Shape : 0.5
Compactness : 0.5

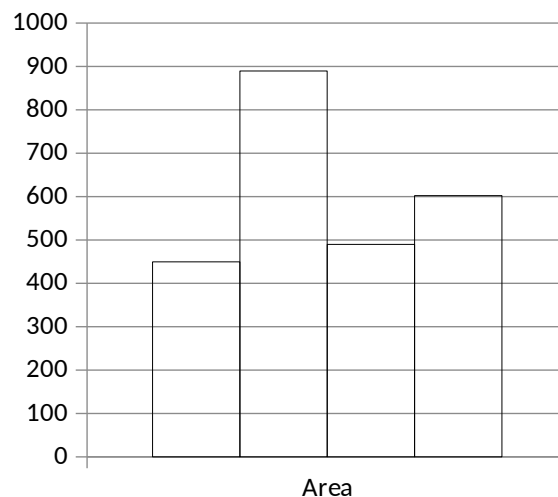
2nd Segmentation
Scale Parameter : 30
Shape : 0.8
Compactness : 0.8

<< Hierarchical region segmentation Feature >>

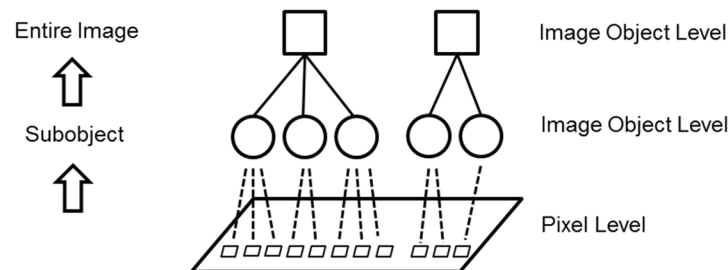
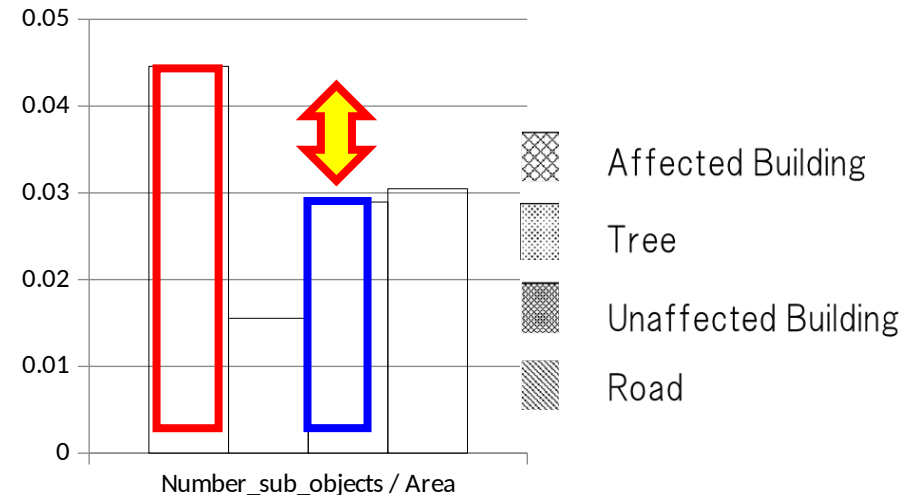
Number_of subobjects



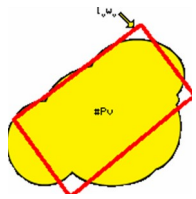
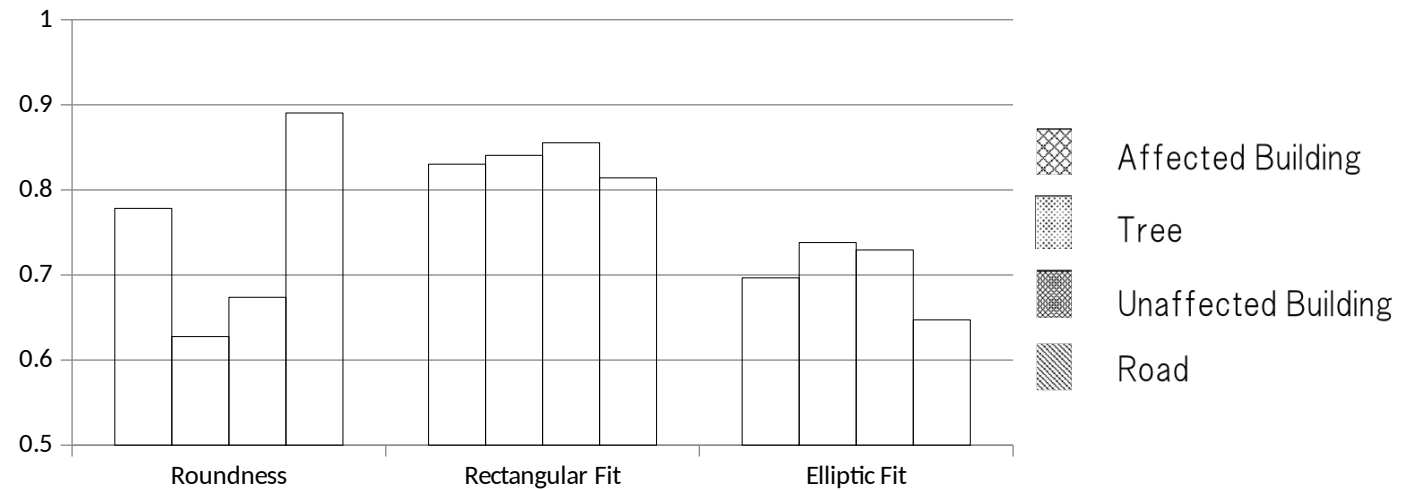
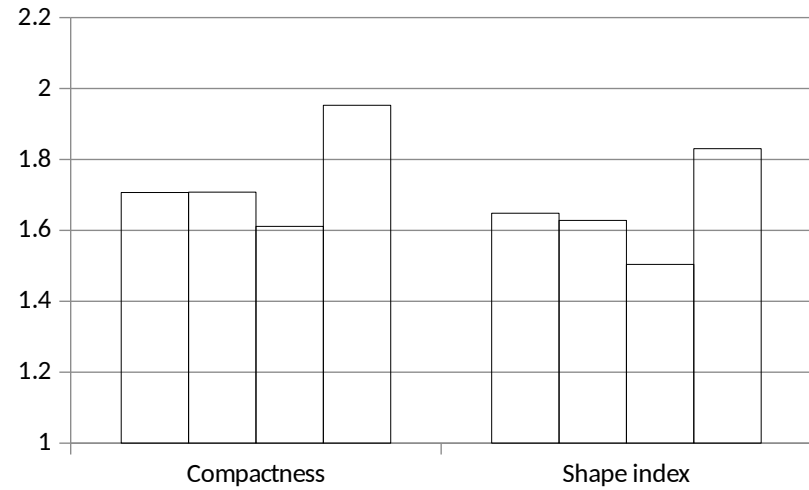
Area



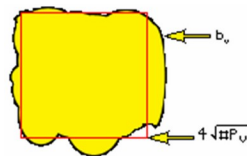
Number_of subobjects / Area



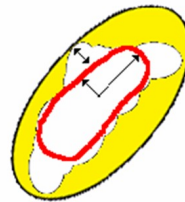
<< Object Shape Feature >>



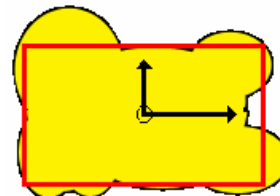
$$\frac{l_v \cdot w_v}{\#P_v}$$



$$\frac{b_v}{4\sqrt{\#P_v}}$$



$$\varepsilon_v \max - \varepsilon_v \min$$

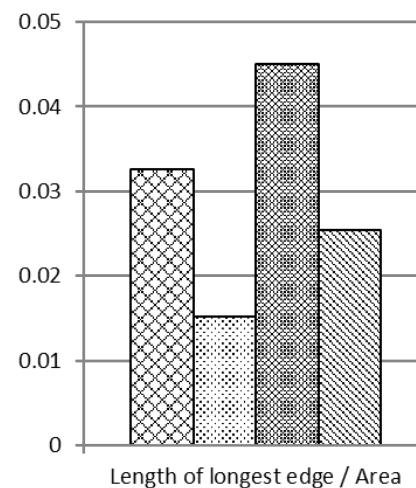
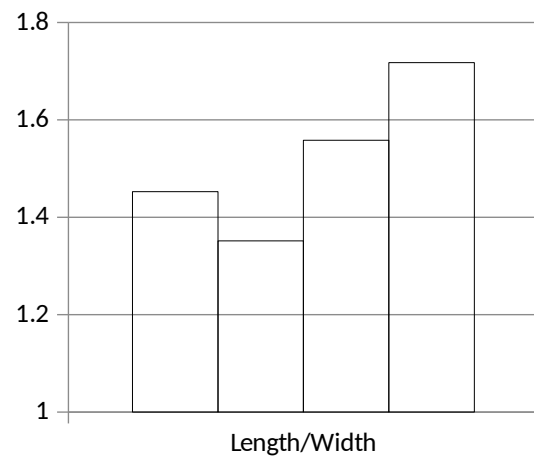
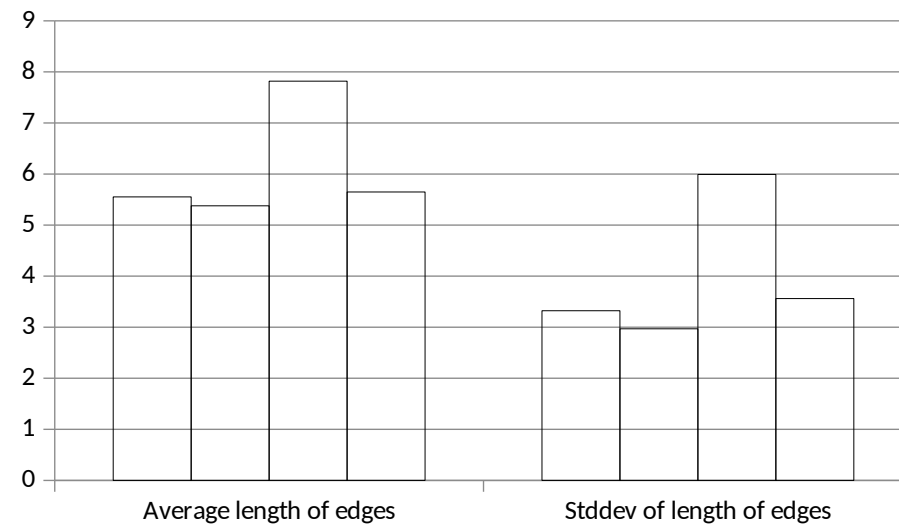
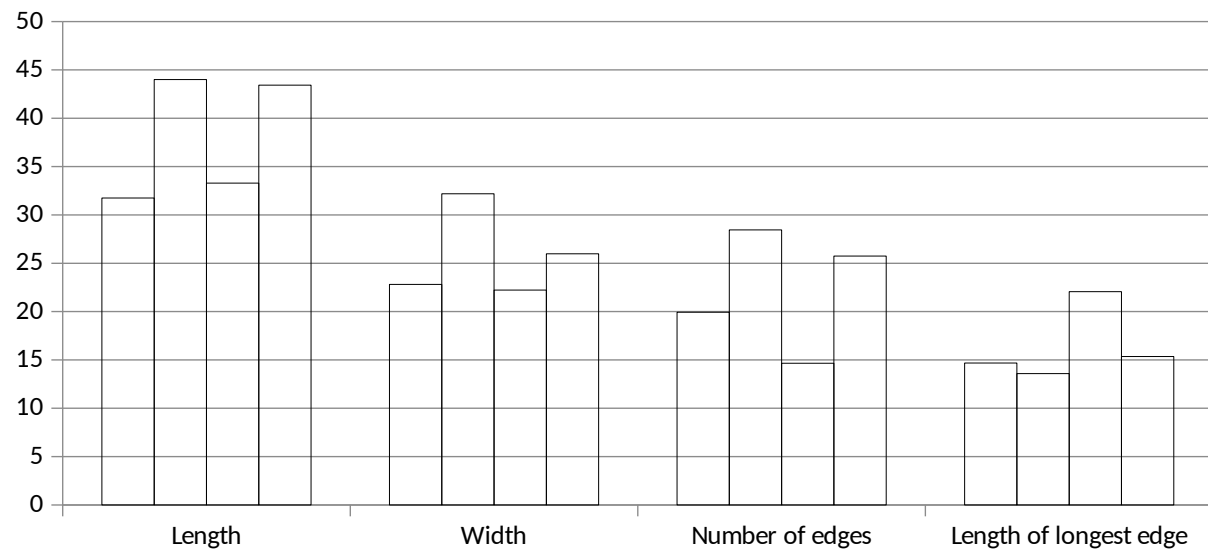


$$\frac{\#\{(x, y) \in P_v : \rho_v(x, y) \leq 1\}}{\#P_v}$$



$$\phi := 2 \cdot \frac{\#\{(x, y) \in P_v : \varepsilon_v(x, y) \leq 1\}}{\#P_v} - 1$$

<< Segmentation Feature >>



<< Useful Features for classifying the Affected Building >>

Damaged Building : 29

Tree : 11
Building : 29
Road : 20

8 Haralick Texture Feature

Homogeneity ,
Stddev length of edges ,
Number of edges
Number of sub-objects

Length ,
Dissimilarity / Homogeneity ,
Number of sub-objects / Area,
Length of longest of edge / Area

Haralick Texture Feature + Hierarchical Feature + Emphasize d Feature

			ID	Classified as		SUM
				1	2	
Class	Deg ree		1	24	5	29
			2	19	41	60
	%		1	82.8	17.2	100
			2	31.7	68.3	100

Accuracy : 73.0%
86.5%

ID	Classified as		SUM
	1	2	
1	26	3	29
2	12	48	60
1	89.7	10.3	100
2	20.0	80.0	100

ID	Classified as		SUM
	1	2	
1	26	3	29
2	9	51	60
1	89.7	10.3	100
2	15.0	85.0	100

83.1%

ID 1 : Affected Building
2 : Unaffected Object (Tree, Building , Road)

Before Earthquake



After Earthquake



Result



Earthquake_damage

0.5 — 0.7

0.7 —

Unextracted locations

$$\text{Earthquake_damage} = -0.036 \times \text{Length} + 3.3368 \times (\text{Dissimilarity/Homogeneity}) \\ + 61.950 \times (\text{Number of sub-objects/Area}) - 36.186 \times (\text{Length of longest of edge/Area}) - 2.506$$

<< Result of verification data >>

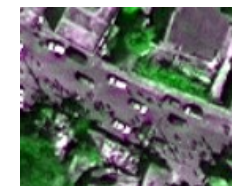
	Result (Number of Objects)		
	Affected Building	Unaffected Object	Sum
Affected Building	67	19	86
Road	5	16	21
Tree	1	104	105
Unaffected Building	7	80	87
	Result(Number of Objects)		
	Affected Building	Unaffected Object	Sum
Affected Building	67	19	86
Unaffected Object	13	200	213

Result (%)		
Affected Building	Unaffected Object	Sum
81.7	18.3	100.0
23.8	76.2	100.0
1.0	99.0	100.0
8.0	92.0	100.0

Result(%)		
Affected Building	Unaffected Object	Sum
81.7	18.3	100
6.1	93.9	100

Total accuracy: 89.3%

Misclassified



Summary

- The purpose of this study was the production of a map to support emergency relief and rescue efforts in a disaster affected area by identifying damage buildings in a short period of time using only a single image of the affected area.
- Object Based Image Analysis
 - **Haralick Texture Feature** : Dissimilarity, Homogeneity
 - hierarchical Feature : Number of Sub-objects, Area
 - **Shape Feature** : Length, Length of longest of edge

$$\text{Earthquake_damage} = -0.036 \times \text{Length} + 3.3368 \times (\text{Dissimilarity/Homogeneity}) + 61.950 \times (\text{Number of sub-objects/Area}) - 36.186 \times (\text{Length of longest of edge/Area}) - 2.506$$

- Sixty-seven out of 86 damaged buildings (81.7%) were identified accurately while 19 damaged buildings (18.3%) were identified inaccurately.

Future Work

- In this research, we conducted the extraction of affected buildings against the earthquake of Republic of Haiti.
- The disaster situation due to the earthquake is characterized by its local culture and economy.
- Therefore, it is necessary to investigate the feature amount effective for earthquake affected buildings in areas with different disaster patterns.



Province



Niigata Province



Sichuan Province



Italy



Chile



Tibet



